

Stochastic modeling of the energy supply system with uncertain fuel price – A case of emerging technologies for distributed power generation

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ABSTRACT

A deterministic energy supply model with bottom-up structure has limited capability in handling the uncertainties. To enhance the applicability of such a model in an uncertain environment two main issues have been investigated in the present paper. First, a binomial lattice is generated based on the stochastic nature of the source of uncertainty. Second, an energy system model (ESM) has been reformulated as a multistage stochastic problem. The result of the application of the modified energy model encompasses all uncertain outcomes together and enables optimal timing of capacity expansion. The performance of the model has been demonstrated with the help of a case study. The case study has been formulated on the assumption that a gas fired engine competes with renewable energy technologies in an uncertain environment where the price of natural gas is volatile. The result of stochastic model has then been compared with those of a deterministic model by studying the *expected value of perfect information (EVPI)* and the *value of stochastic solution (VSS)*. Finally the results of the sensitivity analysis have been discussed where the characteristics of uncertainty of the price of fuel are varied.

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1. Introduction

Energy models are analytical tools for evaluating the behavior of energy systems. They are considered as means for analysis of optimal allocation of resources in the energy sector. A comprehensive review of energy models by Jebaraj and Iniyar [1], Van Beek [2] and Urban et al. [3] indicates that there are various classes of energy models. The models differ from each other in their scope and applications, as well as the techniques that are used for optimizing the development of energy systems.

Energy models with bottom-up structure are associated with “technology paradigm”. Such models are used to identify the optimal mix of technologies that would lead to the minimum of present value of total costs.

When a deterministic energy supply model is applied under the uncertainty, it is rather difficult to obtain a robust result. Therefore, scenario analysis is usually applied for consideration of the impact of uncertain parameters on the results of the deterministic model. In a recent work, Axelsson et al. [4] have presented a tool for creating consistent energy market scenarios adopted for evaluation of energy related investments in the energy intensive industrial processes. Cai et al. [5] have developed an interactive decision

support system for aiding the policy making under uncertainty where a variety of scenarios have been identified based on different energy management policies in a region.

The scenarios might be used to study not only the effects of an energy-environmental policy, but they may also avail themselves to sensitivity analysis through comparison of various scenarios with the base case. Mallah and Bansal [6] have used a parametric sensitivity analysis of uncertain parameters such as price of primary fuels or discount rate in MARKAL energy model. Although approaches based on scenario analysis have been proved to be helpful, but they have had limitation and draw backs when applied in deterministic models. A major shortcoming of scenario analysis is due to the fact that it rarely captures the integration of possible outcomes of the uncertainties.

Further review of literature on optimization under uncertain conditions indicates that stochastic programming is considered as an appropriate approach for handling uncertainties in long term strategic planning [7]. Stochastic programming with a two-stage form has had several applications in energy models [8–11]. Further studies have extended the application of stochastic programming to sequential resolution of uncertainties in a multistage framework. Kanudia and Loulou [12] have applied a three stage stochastic extension of the MARKAL model to study greenhouse gas abatement. They have investigated the advantages of stochastic vs. deterministic models but the optimal timing of capacity expansion based on recourse actions has hardly been discussed.

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Goel and Grossmann [13] have developed a multistage stochastic energy model for optimizing the investment and operational planning of gas field development. They have assumed that the scenario trees would be dependent on decision with a specific probability.

Application of stochastic programming in bottom-up energy models is rather limited. Most and Keles [14] have presented an overview of the latest stochastic models especially dealing with price risks in electricity markets. Further, Krey et al. [15] have developed a framework based on a multi-regional bottom-up energy model. This model includes the stochastic risk function and investigates the impact of uncertain energy prices on the supply structure of energy system. Also Krey and Riahi [16] have developed a decision modeling framework that takes into account the effect of risks. Their risk-averse approach is applied with the help of stochastic version of MESSAGE and it integrates stochastic risk function into a deterministic model. The approach adopted by Krey and Riahi has enabled both avoiding computational complexities and smooth transition from deterministic to a stochastic modeling. They have used the mean value of the uncertain parameters as part of the objective function of a deterministic model. Therefore, the value of the stochastic solution has rarely been estimated. It is shown in the present paper that the value of the stochastic solution would be considerable when a deterministic bottom-up energy model is reformulated as a multistage stochastic model with recourse. In addition, the integrated consideration of uncertainties in the bottom-up energy model would be a distinct feature of the present work.

In order to demonstrate the concept and the applicability of aforementioned approach, a case study has been introduced based on the development of a real energy system where price of primary energy, namely, the natural gas, would be an uncertain parameter. The uncertainty in the price of natural gas is taken into account with the help of the concept of Geometric Brownian Motion (GBM) and the procedure is discussed in Section 2. This approach enables generation of probabilistic scenarios based on the methodology of binomial lattice. Further extension of the methodology towards developing a multistage stochastic problem and its potential to estimate the optimal timing of capacity expansion is discussed in Section 3. The application of the technique is presented in Section 4 where a case study is formulated on the basis of an off-grid distributed power generator. This system would include a parallel arrangement of different technologies which are assumed to supply the required final energy. The problem would then be to identify the optimal path of capacity expansion of the competing technologies under the uncertain price of primary energy. The results of application of the present approach are then compared with the output of a deterministic energy model in Section 5. Finally, the last section of the present paper is devoted to the conclusion.

2. Uncertainty in energy prices

The exogenous parameters of an energy system model include among the others the prices of energy carriers in the international market. In order to demonstrate the application of the approach for handling the uncertainties in the present work, the variation in the price of natural gas would be discussed first. Uncertainty in the price of natural gas stems from the volatility of price of crude oil. In addition to the volatilities in oil price, other variables, such as climate and changes in economy, might also be the sources of volatility in the price of natural gas [17,18]. Studies show that these volatilities in the price of natural gas could be explained by the principles of stochastic processes [17,18]. The most applied stochastic processes for prices of natural commodities include those

of Geometric Brownian Motion (GBM) and mean reverting processes. A review of the characteristics of these processes indicates that the price of natural gas in the long-run could be simulated by mean-reverting models in which the price tends to approach to an average level reflecting its long-run marginal production cost. On the other hand, it has been pointed out that the concept of GBM could provide an appropriate method for handling short and medium term uncertainty in the price of natural gas [19–21]. Analytical solution of the relevant stochastic differential equation of GBM ($dP = \mu \cdot P \cdot dt + \sigma \cdot P \cdot dW$) concludes a discrete time interval for the price of gas as follows [19]:

$$P_t = P_{t-1} \cdot e^{\left\{ \left(\mu - \frac{\sigma^2}{2} \right) \cdot \Delta t + \sigma \cdot \sqrt{\Delta t} \cdot \omega_t \right\}} \quad (1)$$

P_t represents the random asset price at time t , ω is the underlying uncertainty driver as a random number with unit standard deviation and zero mean. μ is drift coefficient and σ is the volatility. This approach has been frequently used for stochastic dynamic problems which are solved by Monte-Carlo simulation and Bellman backward methods. But direct application of GBM in bottom-up energy models based on linear programming is limited [14,22,23]. Based on formulation (1), the sequential realization of the random price of natural gas will lead to a multistage programming framework that could be solved by introducing the probabilistic scenarios [24].

In order to generate the discrete probabilistic scenarios based on the GBM, the method of Cox–Ross–Rubinstein has been used in finance [25]. Geometric Brownian Motion follows the random walk characteristics and its behavior resembles the application of Cox–Ross–Rubinstein method. Applying the Cox–Ross–Rubinstein method and using the characteristic coefficients of GBM, i.e. (μ , σ , Δt) may lead to the development of the binomial lattice. Having the value of price and probability of any node at time step t , the price and probability of next upward and downward nodes based on risk-neutrality assumption are obtained by the following measures [19]:

$$\begin{aligned} \text{Price at the next upward node : } P_{t+1}^u \\ = P_t \cdot (e^{\sigma \cdot \sqrt{\Delta t}}) \text{ with probability multiplier prob} = \frac{e^{\mu(\Delta t)} - e^{-\sigma \cdot \sqrt{\Delta t}}}{e^{\sigma \cdot \sqrt{\Delta t}} - e^{-\sigma \cdot \sqrt{\Delta t}}} \end{aligned} \quad (2)$$

$$\begin{aligned} \text{Price at the next downward node : } P_{t+1}^d \\ = P_t \cdot (e^{-\sigma \cdot \sqrt{\Delta t}}) \text{ with probability multiplier } 1 - \text{prob} \end{aligned} \quad (3)$$

This concept generates a binomial lattice that approximates the GBM in a discrete space and produces the probabilistic scenarios (Fig. 1). For adjusting the Δt , it seems annual time step would be optimal in the present case study which has been frequently used in other research works [22,23].

Fig. 2 summarizes the process of modeling the uncertainties and generating the scenarios as a module which is linked to the energy model.

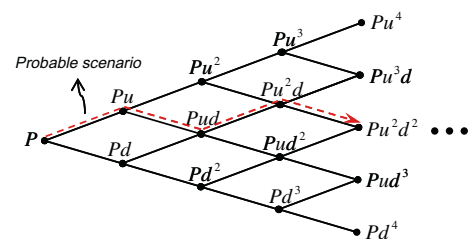


Fig. 1. Generation of the probabilistic scenarios based on binomial lattice.

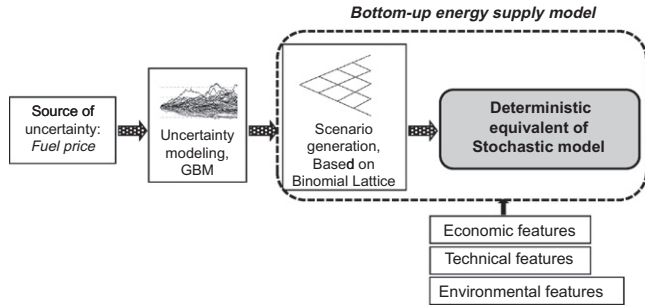


Fig. 2. Interaction between module of scenario generation and energy system model.

3. Stochastic vs. deterministic models

In order to analyze the impact of fuel price volatility on the optimization of energy system, the deterministic energy model of ESM (historical background and description of ESM is described in Appendix A) is developed as a stochastic programming problem in the present paper.

The objective function of the present deterministic energy system model is the minimization of discounted total costs, which may be presented according to the relationship (4),

$$Z = \text{Min} \sum_i \left(\underbrace{\sum_{t=1}^T \frac{C_i \cdot y_{i,t}}{(1+r)^t}}_{\text{Capital cost}} + \underbrace{\sum_{t=1}^T \frac{M_{i,t} \cdot x_{i,t}}{(1+r)^t} + \sum_{t=1}^T \frac{P_{i,t} \cdot x_{i,t}}{(1+r)^t}}_{\text{Maintenance \& Operation Cost}} \right) \quad (4)$$

where Z is the objective function; C the capacity build-up cost (\$/kW); M the maintenance cost (\$/kW); P the fuel price (\$/kW); y the capacity build-up (kW); x the consumed fuel (kW); r the discount rate; i the technology type and t is the time period (Year). The main included constraints in the energy system model are as follows,

Demand constraint:

$$\sum_i (x_{i,t} \cdot \eta_i) \geq \text{Dem}_t \quad (5)$$

where Dem is the energy demand (kW); η the end-use technology efficiency and i is the technology type.

Capacity constraint:

$$\frac{x_{i,t} \cdot \eta_i}{\Delta t} \leq \sum_{\tau=1}^t y_{i,\tau} + H_i \quad (6)$$

where H is the capacity that has been built before base time (kW).

Resource constraint:

$$R_{i,t} \leq R_{i,0} - \sum_{\tau=1}^{t-1} x_{i,\tau} + \sum_{\tau=1}^t D_{i,\tau} \quad (7)$$

R_0 is the initial reserve of resource (kW); R the reserves of exhaustible resource (kW); x the production of primary energy (equal to flow rate of fuel) (kW); D the discovery of new reserves (kW) and i is the resource type.

Uncertainties in certain parameters of an energy supply model could be handled with the help of three different approaches. The main features of three approaches are discussed as follows:

3.1. Wait-and-see (WS)

This common approach is usually based on application of a deterministic model in the framework of various scenarios. Out-

come of each scenario is then attributed a probability of occurrence. Such a process of estimation may be summarized according to the relationship (8).

$$Z_{WS} = \text{Exp}(Z_{\text{scenarios}}) = \sum_i \text{Pr}_i \cdot Z_i \quad (8)$$

where i is the number of scenarios and Z is the value of objective function.

3.2. Expected value (EV)

The second approach has been founded on considering the expected value of the random parameter in a deterministic model (formulation 9).

$$Z_{EV} = \text{Min} \quad Z(\text{Exp}(P)) \quad (9)$$

where P is the price of energy carrier.

3.3. Stochastic model with recourse (RP)

When the random variation of a stochastic parameter, such as energy price, is represented in the model endogenously, an approach based on the stochastic programming is intended. Stochastic programming paradigm with recourse has been basically founded on segregation of the decision variables into two distinct partitions. First stage variables are decided prior to the uncertainty realization. The second stage variables are associated with decisions that have considered uncertainty realization. Since the coefficients of the objective function in the present model are subject to the uncertainty; the first stage might be implemented to identify the decisions on the investment while the second stage would be used to evaluate the impact of variation in the parameters related to the operation.

Generally a two-stage stochastic model could be formulated according to the expression (10).

$$\text{Min}_{x_1 \in X_1} f_1(x_1) + \text{Exp}[\text{Min}_{x_2 \in X_2(x_1, \omega_2)} f_2(x_2, \omega_2)] \quad (10)$$

where

$$f_1(x_1) := c_1^T \cdot x_1, X_1 := \{x_1 : A_1 \cdot x_1 = b_1, x_1 \geq 0\}, \\ f_t(x_t, \omega_t) |_{t=2} := c_t^T \cdot x_t, X_t(x_{t-1}, \omega_t) |_{t=2} := \{x_t : B_t \cdot x_{t-1} + A_t \cdot x_t = b_t, x_t \geq 0\} \quad (11)$$

In a quasi-dynamic energy model the resolution of uncertainties appear sequentially at each period. Therefore, further extension of a two-stage model to a multistage stochastic model shall be required. In a generic form, a T -stage stochastic programming problem can be written in the form of nested formulation which would be expressed as (12) [26]:

$$\text{Min}_{x_1 \in X_1} f_1(x_1) + \text{Exp}[\text{Min}_{x_2 \in X_2(x_1, \omega_2)} f_2(x_2, \omega_2) + \text{Exp}[\dots + \text{Exp}[\text{Min}_{x_T \in X_T(x_{T-1}, \omega_T)} f_T(x_T, \omega_T)]]] \quad (12)$$

$x_t \in R^{n_t}$, $t = 1, \dots, T$ are decision variables and $X_t : R^{n_{t-1}} \times R^{d_t} \Rightarrow R^{n_t}$, $t = 2, \dots, T$ are measurable closed valued multi-functions. It is assumed that data in the initial stage are deterministic in nature.

At the next level the probabilistic scenario trees, which are based on binomial lattice, shall be introduced to formulate the deterministic equivalent form of the multistage stochastic model. Construction of an equivalent deterministic model would be based on the important characteristics of the scenario trees i.e. *Non-anticipativity*. Non-anticipativity constraint stresses that any decision variables with the same historical scenario path should be the same. Deterministic equivalent of the multistage stochastic energy model could be summarized in relationships (13).

$$Z_{RP} = \text{Min} \sum_j \text{prob}_j \cdot \left(\underbrace{\sum_i \sum_{t=1}^T \frac{C_i \cdot Y_{i,t,j}}{(1+r)^t}}_{\text{Capital cost}} + \underbrace{\sum_i \sum_{t=1}^T \left\{ \frac{M_{i,t,j} \cdot X_{i,t,j}}{(1+r)^t} + \frac{P_{i,t,j} \cdot X_{i,t,j}}{(1+r)^t} \right\}}_{\text{Annual operating cost}} \right) \quad (13)$$

where prob is the probability and j is the scenario number, $j \in \{1, 2, \dots, k\}$.

Demand constraint

$$\sum_i (X_{i,t,j} \cdot \eta_i) \geq \text{Dem}_t \text{ for each } (j)$$

Capacity constraint:

$$\frac{X_{i,t,j} \cdot \eta_i}{\Delta t} \leq \sum_{\tau=1}^t Y_{i,t,j} + H_i \text{ for each } (j)$$

Resource constraint:

$$R_{i,t,j} \leq R_{i,0} - \sum_{\tau=1}^{t-1} X_{i,\tau,j} + \sum_{\tau=1}^t D_{i,\tau} \text{ for each } (j)$$

In the present work, data related to the binomial lattice are directly merged into the constraints of the deterministic equivalent problem; hence explicit consideration of the non-anticipativity would be avoided. The generated stochastic problem may then be solved with the help of LP optimizer.

3.4. Optimal timing in multistage stochastic model

The formulated multistage stochastic energy model with recourse is categorized as an LP system. This model optimizes the present value of the total costs of capacity expansion of energy system under the uncertain fuel price. The subject of capacity expansion in energy sector may be explained as optimal timing for investment under uncertainty. This issue has been extensively discussed by the Real Option theory. The concept of Real Option has been elaborated in detail by Dixit and Pindyck (1994) [21] as a new approach for irreversible capital investments under uncertainty. Theory of Real Option states that by assuming flexibility on delaying a decision on investment and evaluating the postponement as an opportunity cost of the decision in volatile environment, the optimum timing for decisions could be improved [21,27,28].

Models of Real Options are most conveniently translated into a mathematical model that could be used with the help of a dynamic programming formulation. This procedure can hardly be applied in a bottom-up energy model that uses LP optimizer. Cheng et al. [7] has shown numerically, a problem of optimal control is equivalent to a problem of stochastic programming. But the present work emphasizes more specifically that a multistage stochastic approach in an energy model resembles Real Option problem with exogenous rate of interest. Appendix B shows that optimal timing in the multistage stochastic energy model is equivalent to the paradigm of a Real Option.

4. Application of the model

Distributed generation (DG) systems in small and medium sizes have been developed in many countries. Expansion of DG systems has led to further development of private generation and competitive energy markets, reduction in the distribution network losses, increase in the generation system availability, penetration of the

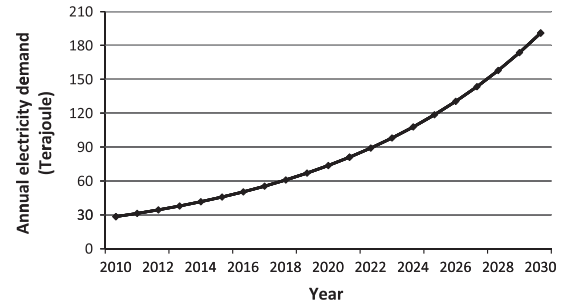


Fig. 3. Demand curve for DG system case study [33].

renewable energies and reduction in the adverse emissions and green house gases [29–31].

Recent studies on the potentials of DG systems in Iran indicate that utilization of gas fired technology may provide the most economic DG systems that will be followed by the wind turbine (WT) and photovoltaic (PV) in remote areas [32]. It has been concluded that a combination of gas fired technology together with the small scale renewable technologies could provide a reliable foundation for the technical and environmental improvement of energy supply system and it may facilitate competition in the energy market.

In order to demonstrate the applicability of the model, the case of penetration and expansion of DG systems in the Iranian national power network has been studied. Based on estimation of power generation authorities [33], it is further assumed that the part of power demand that ought to be met by DG systems in a remote area would follow the trend that could be observed in Fig. 3.

Economical and technical parameters used in the model at base time are listed in Table 1 which represents the local situation in the power system [34–36].

As it was discussed in Section 2, in order to model the GBM behavior of the price of gas, its volatility and drift coefficients shall have been derived from historical data according to the relationship (14).

$$\begin{aligned} \mu(\text{drift}) &= \frac{1}{T} \sum_{t=2}^T \left(\ln \frac{P(t)}{P(t-1)} \right), \quad P = \text{Annual price} \\ \sigma(\text{volatility}) &= \sqrt{\frac{1}{N} \sum_{t=1}^N \left\{ \frac{[P(t_i) - P(t_{i-1})]}{P(t_{i-1})} - \left(\frac{[P(t_i) - P(t_{i-1})]}{P(t_{i-1})} \right) \right\}^2} \end{aligned} \quad (14)$$

The results of calculation for a time period of two decades are indicated in Table 2.

These data are used to generate the coefficients of the binomial lattice as approximations of GBM according to the procedure which was explained in Section 2. Direct formulation of the outcome of binomial lattice in constraints of the model would decrease the size while it includes the non-anticipativity. In order to reduce the number of scenarios within the binomial lattice, further application of scenario reduction algorithms is considered. Hence, a scenario reduction module of the GAMS/SCENRED software has been used based on the fast backward method [37].

5. Results of model application

Penetration and capacity expansion of DG systems in Iranian power network under fuel price uncertainty have been studied with the help of both deterministic energy supply model and multistage stochastic model. The results of both deterministic and stochastic models have been compared with each other in order to demonstrate the potentials of developed stochastic model.

Table 1
Technology specifications based on local information and references [34–36].

	DG technologies		
	Gas engine	PV	WT (Onshore)
Capital cost (\$/kW) (in 2010)	1000	6000	2100
O&M cost (\$/kWyr)	87.6	87.5	210.24
Efficiency (%)	38	17	33
Capacity factor (%)	75	25	26

Table 2
Price coefficients of natural gas represented as GBM (source of prices: BP outlook report 2010).

	Volatility (σ) (%)	Drift (μ) (%)
Gas price	19.88	1.26

5.1. Results of stochastic vs. deterministic models

Results of application of three approaches (*WS*, *EV* and *RP*) shall be compared with each other. To evaluate the *EV* solution, *EEV* is introduced as the expected result of using the solution of the deterministic model *EV* [24]. *EEV* is basically defined for the two-stage models and it is calculated by solving the recourse problem while the first stage decision variables are fixed to the solution of *EV* problem. Regarding the relation between these values, Madansky theorem [24] explains that for a two-stage stochastic program $WS \leq RP \leq EEV$. Escudero et al. [38] have extended this concept for the multistage problems. On this basis, $EEV(t)$ is defined for each stage (t) as the solution of the recourse problem while the decision variables up to stage ($t-1$) should be fixed to the result of deterministic *EV* model. It has been proved that for each stage of the multistage problem $EEV(t) \leq EEV(t+1)$ and so $EEV(T)$ in T stage model could have mean similar to *EEV* in two-stage model. Data from Table 3 confirms the extension of Madansky theorem for the multistage stochastic problem.

It may be observed from Table 3 that the results of all three methods show little differences at the initial stages and initial time points. When time horizon extends further beyond the initial points, divergence of the results between three methods is revealed to be considerable.

The difference between *WS* and *RP* solutions is defined as the *expected value of perfect information (EVPI)* [12,24,39]. It shows the maximum amount a decision maker would be ready to pay in return for complete information of the future (Fig. 4). Such a willingness to pay is increased as time horizon extends beyond the initial stages.

The difference between *RP* and *EEV* solutions is defined as the *value of stochastic solution (VSS)* [24,38,39]. It shows the cost of ignoring the uncertainty in the process of decision making. Fig. 5 shows that the cost of ignoring the uncertainty would be consider-

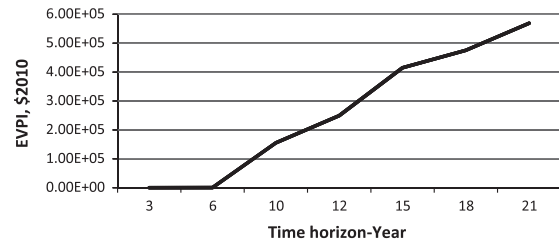


Fig. 4. Trend of expected value of perfect information.

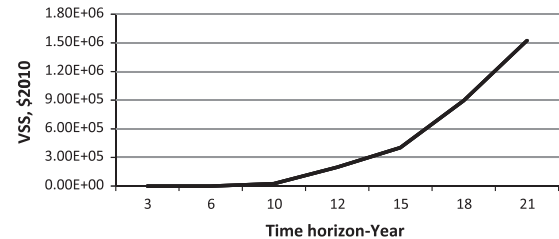


Fig. 5. Trend of value of stochastic solution.

able in the long run. It amounts up to 11.5% of the total cost of the energy supply system in this study.

Comparison of the results shows that *RP* solution could be more robust than *WS* since it encompasses all the uncertain possibilities together. *WS* solution does not reflect the uncertainty resolutions and cannot capture the system bifurcations. The *EV* solution could be sub-optimal since it does not include any flexibility in making decisions under uncertainty. Therefore, it may be concluded that the *RP* solution of the multistage problem could be optimal because it considers advantage of optimal timing.

Analysis of the penetration and expansion of clean technologies, such as wind turbine (WT) and photo voltaic system (PV) under uncertainties involved with the price of fuel, has been the subject of the present case study. Fig. 6 shows the expansion of capacities for three competing technologies. The trend of development shows the optimal time points for introduction of PV and WT technologies.

It may be observed that gas engine technology is the primary choice of the model. But as the capital costs of WT and PV technologies are reduced, an increase in the capacity build up of new technologies is concluded. WT technology seems to be economically marginal but PV needs more improvement.

Table 3
Comparison of the results of different optimization methods.

Time horizon (Yr)	Obj function (\$2010)		
	WS	$\leq RP \leq$	EEV
3	3,567,663	3,567,663	3,567,663
6	5,976,118	5,976,599	5,976,600
10	8,560,527	8,715,717	8,741,595
12	9,766,053	1.00E + 07	1.02E + 07
15	1.10E + 07	1.14E + 07	1.18E + 07
18	1.24E + 07	1.29E + 07	1.38E + 07
21	1.37E + 07	1.42E + 07	1.58E + 07

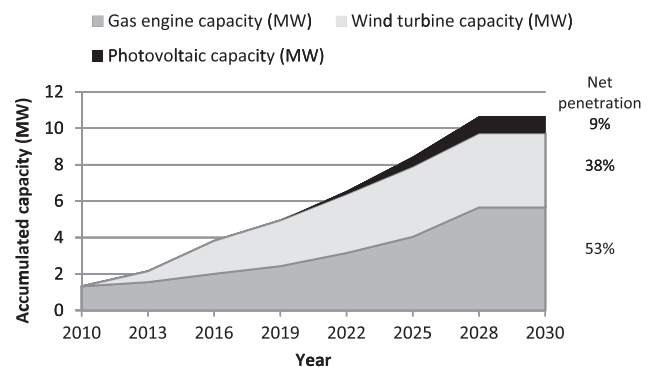


Fig. 6. Expected capacity expansion of competing technologies.

Table 4

Sensitivity analysis of the results of stochastic model with respect to the price volatility of natural gas.

Gas price volatility change (%)	–10	–5	0	+5	+10
Renewable tech. (WT + PV) expected penetration (%)	58.28	57.19	56.09	54.98	53.85

Table 5

Sensitivity analysis of the result of stochastic model with respect to the drift coefficient of the price of the natural gas.

Gas price drift change (%)	–20	–10	0	+10	+20
Renewable tech. (WT + PV) expected penetration (%)	54.17	55.14	56.09	56.88	57.81

5.2. Impact of uncertainty in the price of fuel

The uncertain price of natural gas was supposed to follow a Geometric Brownian Motion with given volatility and drift measures. Table 4 shows the result of sensitivity analyses based on stochastic energy model with respect to the changes in the volatility of the price of natural gas. This shows that when the price of natural gas is fluctuated, less investment on renewable technologies is preferred. As the volatility in the price of fuel is increased, there would be higher opportunity costs of waiting for investment on costly renewable technologies until next period. On the other hand, smaller degrees of uncertainty lead to optimal timing in investments by rapid expansion and diffusion of renewable energy technologies.

Sensitivity analysis on the drift coefficient of the price of fuel (Table 5) shows that, any growth of the rate of price of natural gas would cause an increase in operational cost of gas engine technology which may lead to high economic feasibility of renewable energy technologies.

6. Conclusion

To enhance the applicability of a deterministic energy supply model in an uncertain environment, a method of integrated consideration of the uncertainties and the process of developing the multistage stochastic energy model have been investigated in the present paper. First, in order to explicit consideration of uncertainty in the model, the stochastic characteristic of the random parameters is simulated as a Geometric Brownian Motion (GBM). The GBM has been approximated by Cox–Ross method in order to generate probabilistic scenario trees and a binomial lattice. Second, a multistage stochastic model has been formulated based on the deterministic energy supply model. The lattice is then used as an element of the multistage stochastic energy model. This model encompasses all the uncertain outcomes and enables optimal decisions. The comparison of the results with those of deterministic approach i.e. wait-and-see solution and expected value solution shows that the bottom-up stochastic energy model represents the real situation effectively. This stochastic model would have the potential of identifying the optimal timing of investment which is equivalent to a Real Option problem.

Effectiveness of the stochastic energy model has been demonstrated with the help of a case study. The case study has been formulated on the assumption that a gas fired engine in an off-grid DG system competes with renewable energy technologies in an uncertain environment where the price of natural gas is volatile. Calculation of EVPI as the cost of information on uncertainty and VSS as the cost of ignoring the uncertainty shows that

the performance of a stochastic energy model is enhanced as the time horizon is extended. The results of the application of the model indicate that the competition between gas engine, wind turbine and photovoltaic is affected by changes in the prices of natural gas considerably. Sensitivity analysis shows that the coefficients of price volatility and drift of natural gas has considerable impact on optimal development of the renewable energy technologies.

Appendix A

Model MESSAGE has been used for analysis of the development of energy system in developing countries in 1980s [40]. Then this model has been regionalized based on the concept of transshipment. The model has also been modified in order to cover the behavior of primary energy fields (such as crude oil and gas fields). The constraints on the exhaustibility of the resources should have been complemented by further constraints based on the type curve of the resources and fields. In addition to the above modifications, the model has also been transformed into a parametric programming where some of exogenous parameters were considered as a function of a specific parameter. Transformation of the model into parametric programming has enabled scenario reduction which has enhanced the applicability of the model. The new version of the energy supply model is called ESM (energy system model) and it has been applied for studying the development of energy system in Iran.

Further application of the deterministic model ESM indicated that there are considerable exogenous parameters in the model that are subject to uncertainties. Application of the model would then be possible through scenario analysis. But, shortcomings of the scenario analysis to find a robust solution motivated the present work in order to extend the model to a multistage stochastic problem in which integrated consideration of the uncertainties would be possible.

Appendix B

If an energy model, with uncertainties within its parameters, is formulated as a dynamic program based on Real Option approach, the objective function DC_t at each time stage t shall be the minimization of expected value of total fixed costs and sum of variable costs of technologies built from that stage to the final period plus the discount of the objective function for the time stage $t + 1$.

$$DC_t = \text{Min Exp} \left\{ \underbrace{\sum_i cfix_{i,v=t} y_{i,v=t}}_{\text{Capital costs}} + \underbrace{\sum_i \sum_{z=t}^T cvar_{i,z,t} (1+r)^{-(z-t)} x_{i,z,t}}_{\text{Variable costs}} + \underbrace{\frac{1}{1+r} \text{Exp}\{DC_{t+1}\}}_{\text{Continuation value}} \right\} \quad (\text{B.1})$$

where; $x_{i,z,t}$: the energy flow consumed by technology i at time z ; which has been built-up at time stage t ; $y_{i,t}$: the capacity build-up for technology i at time stage t ; $cvar_{i,z,t}$ the variable (operating) cost in time z for technology i ; which has been built-up at time stage t ; $cfix_{i,t}$: the fixed (capital) cost in time t for capacity build-up of technology i and r exogenous rate of discount.

If the problem is reformulated in order to include all the time intervals simultaneously, the objective function may be represented as following relationship.

$$\begin{aligned}
Z = \text{Min Exp} \{ & \sum_i cfix_{i,v=1} y_{i,v=1} + \sum_i \sum_{z=1}^T cvar_{i,z,v=1} \\
& (1+r)^{-(z-1)} x_{i,z,v=1} + \dots + \frac{1}{1+r} \text{Min Exp} \{ \sum_i cfix_{i,v=t} y_{i,v=t} \\
& + \sum_i \sum_{z=t}^T cvar_{i,z,v=t} \\
& (1+r)^{-(z-t)} x_{i,z,v=t} + \dots \} + \dots \} \quad (B.2)
\end{aligned}$$

setting the following notations,

$$\begin{aligned}
f_1 &= \sum_i cfix_{i,v=1} y_{i,v=1} \\
f_t &= [\sum_i cfix_{i,v=t} (1+r)^{-(v-1)} y_{i,v=t} \\
& + \sum_i \sum_{\tau=1}^{t-1} cvar_{i,z=t-1,v=\tau} (1+r)^{-(z-1)} x_{i,z=t-1,v=\tau}], \quad t = 2, \dots, T \\
f_{T+1} &= [\sum_i \sum_{\tau=1}^T cvar_{i,z=T,v=\tau} (1+r)^{-(z-1)} x_{i,z=T,v=\tau}] \quad (B.3)
\end{aligned}$$

then the objective function becomes:

$$Z = \text{Min } f_1 + \text{Exp}[\text{Min } f_2 + \text{Exp}[\dots + \text{Exp}[\text{Min } f_{T+1}]]] \dots \quad (B.4)$$

Such a restructuring shows that a dynamic programming formulation based on Real Option approach is equivalent to the nested form of a multistage stochastic problem that could be solved with the help of an LP solver.

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